

**Volve
15/9-F-11 B
USIT/CBL
9 5/8" casing
6 June 2013
Cement Log
Interpretation Report**

Title:		
Volve 15/9-F-11 B USIT/CBL 9 5/8" casing 6 June 2013 Cement Log Interpretation Report		
Document no. :	Contract no.:	Project:

Classification: Internal	Distribution: Corporate Statoil
Expiry date: 2014-01-11	Status Final

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Author(s)/Source(s): Kjersti Hevrøy	
Subjects: Cement Bond Evaluation.	
Remarks:	
Valid from:	Updated:
Responsible publisher: Kjersti Hevrøy	Authority to approve deviations:

Techn. responsible (Organisation unit): DPN OE OSE PTC MP	Techn. responsible (Name): Kjersti Hevrøy	Date/Signature:
Responsible (Organisation unit):	Responsible (Name):	Date/Signature:
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Approved by (Organisation unit): TPD TEX PTEC PT	Approved by (Name): Linda Storetvedt Lien	Date/Signature: 2015-01-07

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Responsible (Organisation unit):	Responsible (Name):	Date/Signature:
Recommended (Organisation unit): TPD TEX PTEC PT	Recommended (Name): Kevin Constable	Date/Signature: 3/1-15 <i>Kevin Constable</i>
Approved by (Organisation unit): TPD TEX PTEC PT	Approved by (Name): Linda Storetvedt Lien	Date/Signature: 2015-01-07 8/1-15 <i>Linda Storetvedt Lien</i>

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1 General Information

1.1 Well Information

Country	Norway	
Field Name	Volve	
Well Name	15/9-F-11 B	
Rig Name	Maersk Inspirer	
Mud Details		
Mud Weight (SG)	1.28	
Mud Type	OBM Enviromul	
Clean out run before logging (after cementing)	Yes	
Casing Size Logged (#1)	9 5/8"	
From mMD	629.2	
To mMD	3192.5	
Casing Weight (lb/ft)	53.5	
Nominal ID (in)	8.535	
Nominal Thickness (in)	0.545	
Previous Casing Shoe Depth	2570.7	14" casing
Max Deviation in Logged Interval	63.5	

1.2 Logging Objectives

- Evaluate quality of cement in relation to requirements for placement of production packer
- Identify TOC

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1.3 Cement Job Information

Cement Job Date (and time if logging <48hrs after cementing)	04.06.2013	<i>Cement displacement completed 05:00</i>
Cementing Details		
Cement Type	Class G	
Cement Density	1.9 GTTTZ	
Cement Volume Pumped	16.8 m3	
Expected Cement Impedance at time of logging	5.6 MRayl	
Theoretical Top of Cement	~2670m MD	<i>With 9 m3 loss 2980m</i>
Mud in the well when cementing (displacing)	1.28 SG OBM	
Cementing Operations		
Losses during cementing	Yes	<i>7-9 m3</i>
Any other issues during cementing operations	No indication of bumping top plug	

Report date: 04.06.2013						
Casing size 9 5/8"	Stage cementing: No		Top up job No	Company Halliburton		Contract no
Theoretical TOC MD 2 792,0 m	Evaluated TOC MD m	Evaluated by	Liner rotation planned No	Liner rotation achieved		
Remarks 10 3/4" x 9 5/8" casing						
Fluids pumped	Type	Density g/cm3	Volume m3	Pump rate l/min	Pump pressure bar	Return
Spacer before	Cement Slurry GTTTZ	1,52	28,0	1 200	40,00	Full
Tail	Cement Slurry GTTTZ	1,90	16,9	650	35,00	Partiel
Compositions						
Fluids pumped	Component	Quantity	Unit	Density g/cm3	Premixed in water	
Spacer before	BARITE	755,00	kg/m3	1,39		
	Freshwater	22,00	m3	1,00		
	MUSOL E	0,40	m3	0,92		
	NF-6, DEFOAMER	0,04	m3	0,94		
	SEM-8, WATER WETTING AGENT	0,40	m3	1,16		
	Tuned Spacer E	74,00	sacks			
Tail	CFR-8L	2,00	l/100kg			
	EZ-FLO added to cmt onshore	0,10	l/100kg	1,39		
	Freshwater	45,23	l/100kg	1,00		
	GASCON	5,00	l/100kg	1,10		
	HALAD 400L	4,00	l/100kg			
	HR-5L	1,40	l/100kg	1,16		
	NF-6, DEFOAMER	0,10	l/100kg	0,94		
	Norcem "G"	102,26	kg/100kg	3,22		
	SSA-1	35,00	kg/100kg	2,63		

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1.4 Logging Operations

Logging Passes	Date and Time	From (mMD RKB)	To (mMD RKB)	Log Resolution
Main Pass	05.06.2013 05:24	2493	3180	5 deg 6 in
Repeat Pass	05.06.2013 03:33	2950	3180	5 deg 6 in

Tool String	USIT-CBL-GR-CCL	C-Sub
Tractor Used	No	

2 Log Quality Control

2.1 Fluid Properties Measurement

FVEL Measurement Quality	☒	☐	☐	
ZMUD Measurement Quality	☐	☐	☒	<i>Too high (commonly experienced in OBM)</i>
Measured Fluid Velocity Range in Main Pass Interval (us/ft)	233-235			
Measured ZMUD Range in Main Pass Interval (MRayls)	1.8-1.9			As is normally experienced in OBM, the measured mud impedance was too high and theoretical values (with appropriate K factor) were used (calibrated to expected results in the liquid filled free pipe section)
Theoretical ZMUD based on Measured Fluid velocity (MRayls)	1.67			
Mud Impedance used in final data on which the interpretation is based	1.59			$K = 0.95$

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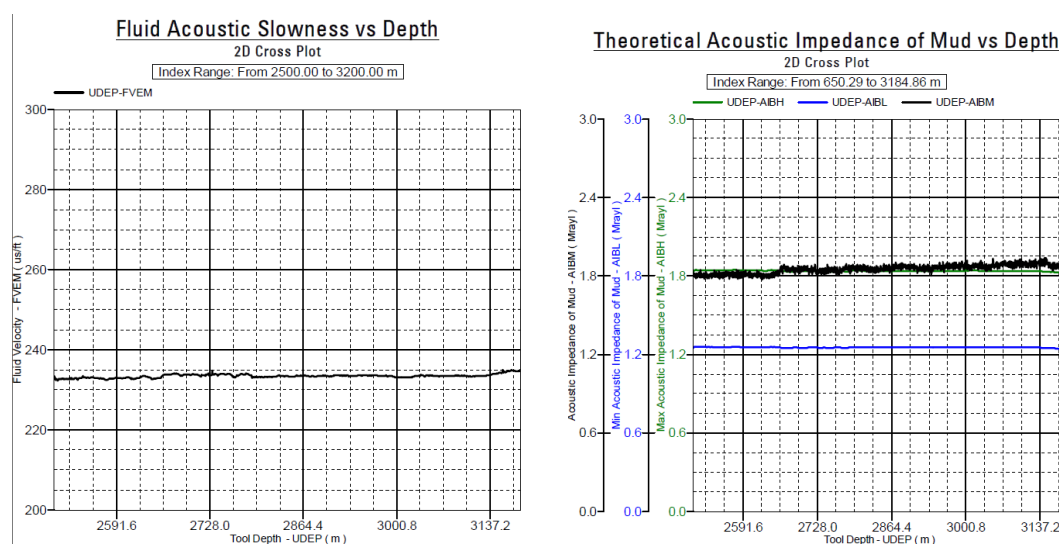


Figure 1: Fluid Properties Crossplots

2.2 USIT Log Quality

Tool Centralisation	☐	☒	☐	<i>Eccentralsation outside tolerance (>2% of csg OD) in large parts of logged interval</i>
First Echo Amplitude	☒	☐	☐	<i>Some slight variation in amplitude levels high side to low side due to tool eccentricalsation, however amplitude levels are still OK</i>
Radius Curves within Specification	☒	☐	☐	
Thickness Curves within Specification	☒	☐	☐	
Manufacturing Patterns visible on thickness image	☒	☐	☐	
Processing Flags	☒	☐	☐	
Repeatability	☒	☐	☐	
Agreement with CBL/VDL	☐	☒	☐	<i>Some differences in some intervals where CBL reads higher than expected based on USIT impedance measurements</i>

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2.3 CBL/VDL Log Quality

Transit Time as expected	☒	☐	☐	
Free Pipe Amplitude as expected	☒	☐	☐	<i>Reference Free Pipe Amplitude = 53mV</i>
Tool eccentricisation (transit time variation)	☐	☒	☐	<i>Some variation in transit times that appear to indicate some tool eccentricisation</i>
Repeatability	☒	☐	☐	

3 Interpreted Results

Please note that in some intervals where the USIT shows well bonded, medium impedance cement the CBL amplitude is not in agreement (too high amplitudes). The CBL is much more sensitive to micro-annulus than the USIT log and given that:

- 1) there is no reason to doubt the USIT data (log quality appears to be very good)
- 2) that the CBL amplitude shows a full dynamic range in the logged interval (the CBL amplitude is consistent with the USIT in the lower section of the log)
- 3) There still appears to be formation arrivals on the VDL in the intervals where the CBL amplitude shows much poorer cement compared to the USIT.
- 4) The mismatch between CBL and USIT often correlates with geological changes.

The mismatch has been attributed to a micro-annulus effect on the CBL. (Probably related to formation effects and related shrinkage effects on the cement) such that the USIT data is the primary

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3.1 Cement Quality

Interval Top (mMD)	Interval Bottom (mMD)	Cement or Formation Bond Quality	Cement or Formation or combination of both	Hydraulic Isolation Expected	If Isolation expected Length of Isolating Interval (m)	Comments
2492	2568	Free pipe	n/a	No		Free pipe in 9 5/8" x14" dual casing annulus
2587	2637	Free pipe	n/a	No		Predominantly free pipe, but with some small patches of poorly bonded formation material (early stages)
2637	2679	Poor	Cement	No		Patchy, relatively low impedance cement with liquid filled areas.
2679	2735	Moderate to Good	Cement	Yes	56	Although there is some variation in impedance around the annulus, the low side, low impedance material appears to be solids and therefore the interval has been interpreted as isolating.
2736	2755	Poor	Cement	No		Very little bonded cement in the interval, this appears to be a geology related effect on the cement as the interval corresponds exactly to a reduction in the GR.
2755	2784	Moderate	Cement	No		Significant bonded cement around the annulus but with a liquid filled channel on the low side of the hole
2784	2836	Moderate	Cement	No		Significant bonded cement around the annulus but with a liquid filled channel which, unusually, is running along the high side of the hole. This high side channel narrows towards the top of the interval.
2836	2845	Moderate	Cement	No		Low side liquid filled channelling
2845	2848	Good	Cement	Yes	3	Well bonded, medium impedance cement around entire annulus
2848	2851	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
2851	2855	Good	Cement	Yes	4	Well bonded, medium impedance cement around entire annulus
2855	2910	Moderate	Cement	No		More variation in cement impedance in the interval. Bonded cement towards the high side but with a consistently low measured impedance on the low side indicative of a continuous fluid filled channel.
2910	2937	Good	Cement	Yes	27	Medium-high impedance cement around entire annulus
2937	2944	Poor	Cement	No		Low side channel
2944	2950	Good	Cement	Yes	6	Well bonded, medium impedance cement around entire annulus

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2950	2956	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
2956	3003	Good	Cement	Yes	47	Well bonded, medium impedance cement around entire annulus
3003	3007	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3007	3009	Good	Cement	Yes	2	Well bonded, medium to high impedance cement around entire annulus
3009	3013	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3013	3016	Good	Cement	Yes	3	Well bonded, medium to high impedance cement around entire annulus
3016	3020	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3020	3040	Good	Cement	Yes	20	High impedance cement around entire annulus
3040	3045	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3045	3127	Good	Cement	Yes	87	Well bonded. medium to high impedance cement around entire annulus. There are a couple of small isolated liquid filled pockets <2m in length but this will not have any impact on overall isolation in the interval
3126	3130.5	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3130.5	3138	Good	Cement	Yes	7.5	Well bonded, medium to high impedance cement around entire annulus
3138	3143	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3143	3145	Good	Cement	Yes	2	High impedance cement around entire annulus
3145	3148.5	Moderate	Cement	No		Significant bonded cement around the annulus but with a wide liquid filled pocket on the low side of the hole
3148.5	3151.5	Good	Cement	Yes	3	Well bonded. medium to high impedance cement around entire annulus
3151.5	3161	Moderate	Cement	No		Significant bonded cement around the annulus but with 2 liquid filled pockets, on the low side of the hole, that extend throughout the majority of the interval.
3161	3180	Good	Cement	Yes	19	Well bonded medium to high impedance cement around entire annulus
Total Cumulative Length (m) of 'barrier quality' cement in logged interval =					286.5	

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Appendix A: Figures

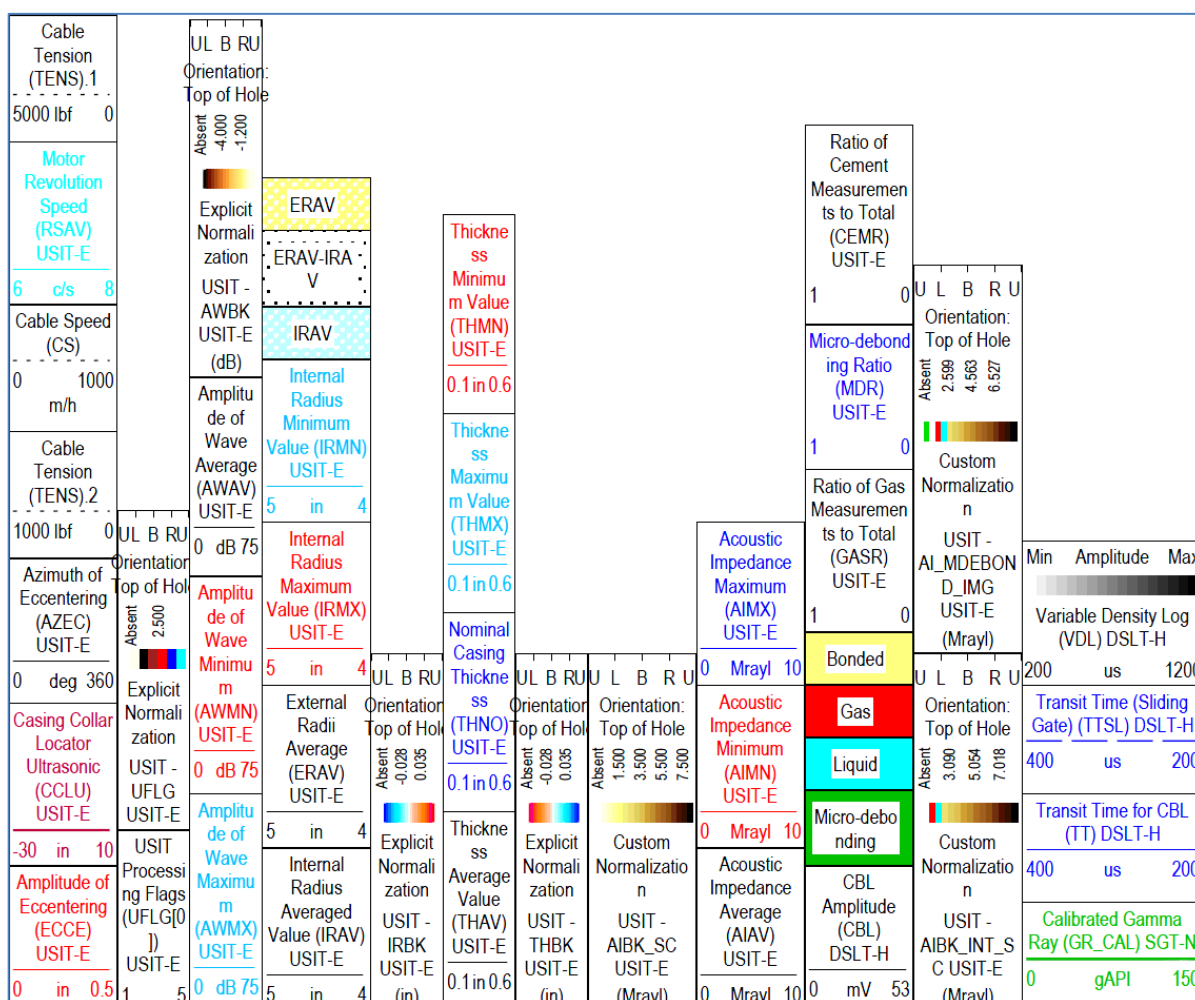


Figure 2: Log header insert for all log images below (showing curves, images, scales etc)

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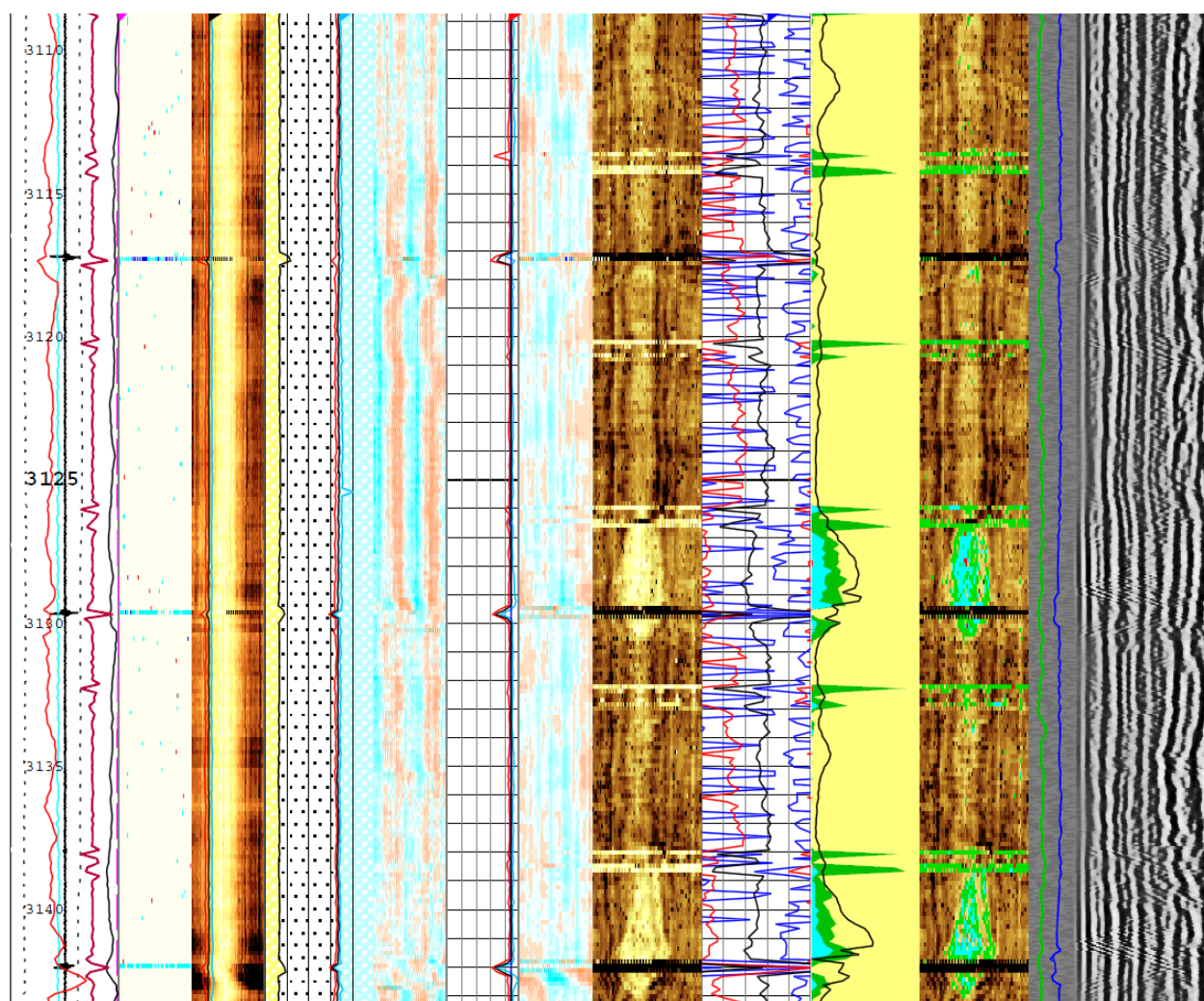


Figure 3: Example log section from 3109-3143m showing predominantly well bonded high impedance cement around the entire annulus, except for some isolated liquid pockets of relatively short extent. Note: the good agreement between the CBL and the USIT.

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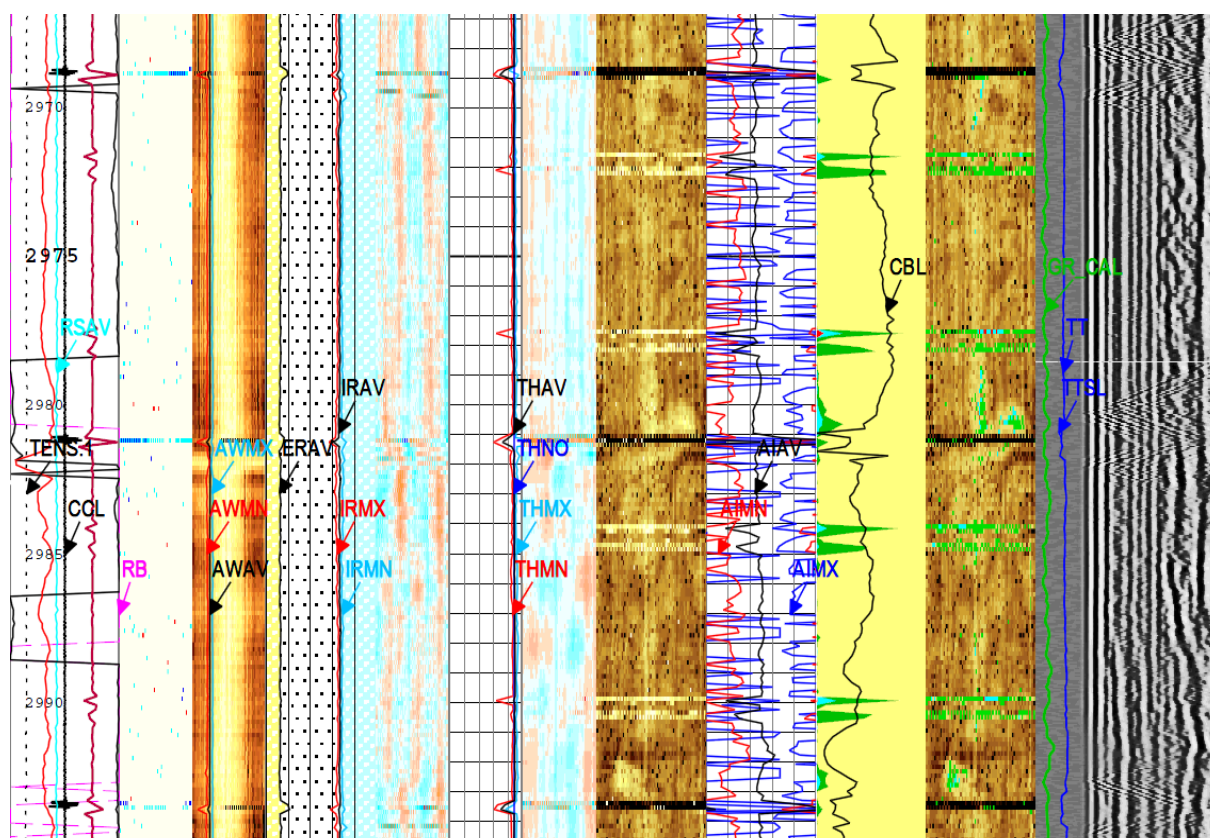


Figure 4: Low section from 2968-2995m where the USIT is showing well bonded, medium to high impedance cement. Note: the CBL amplitude is not in agreement with the USIT and this indicates a probable micro-annulus effect on the CBL amplitude (ref formation arrivals on the VDL even in the presence of a high CBL amplitude)

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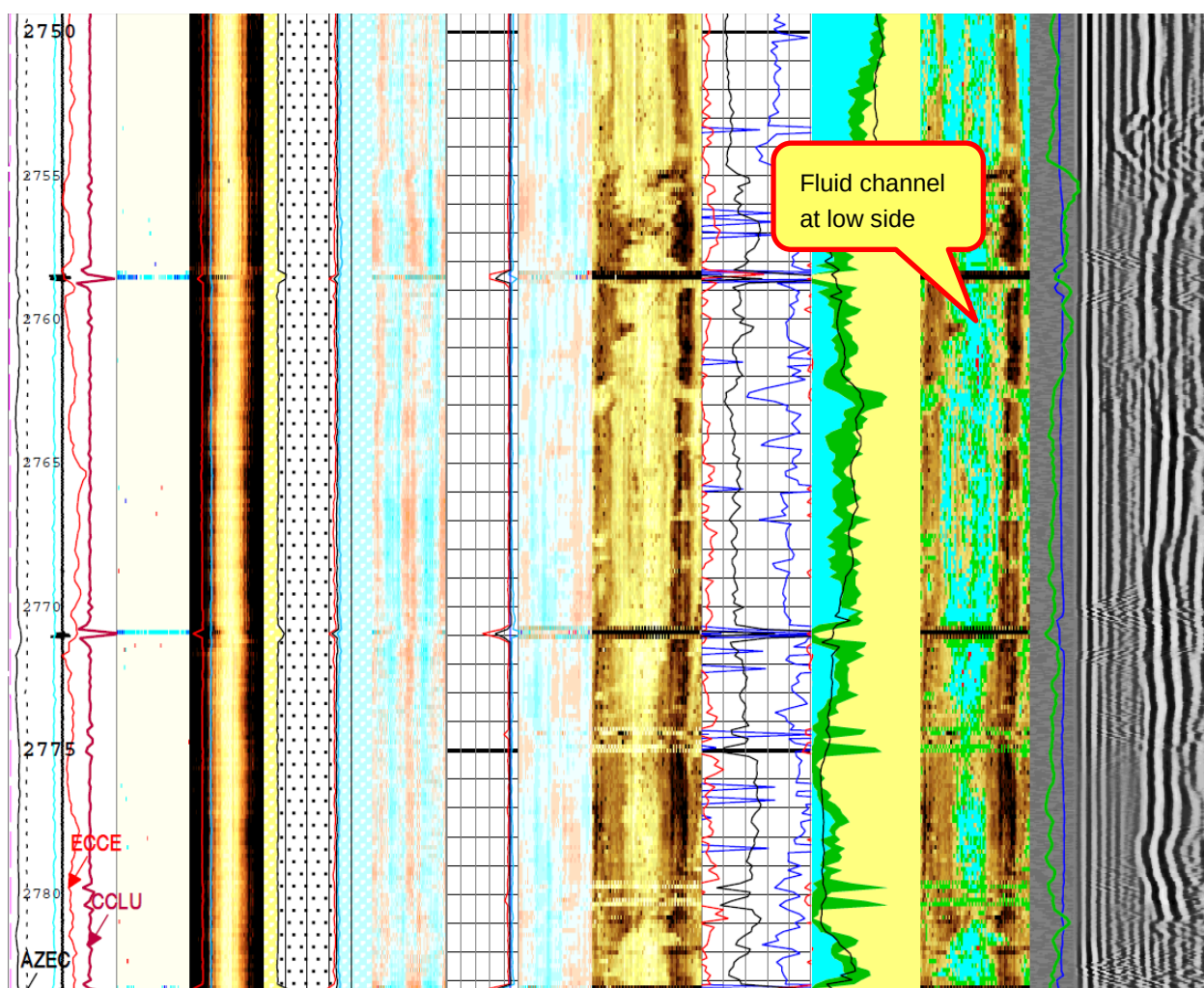


Figure 5: Example Log Section from 2750-2783m

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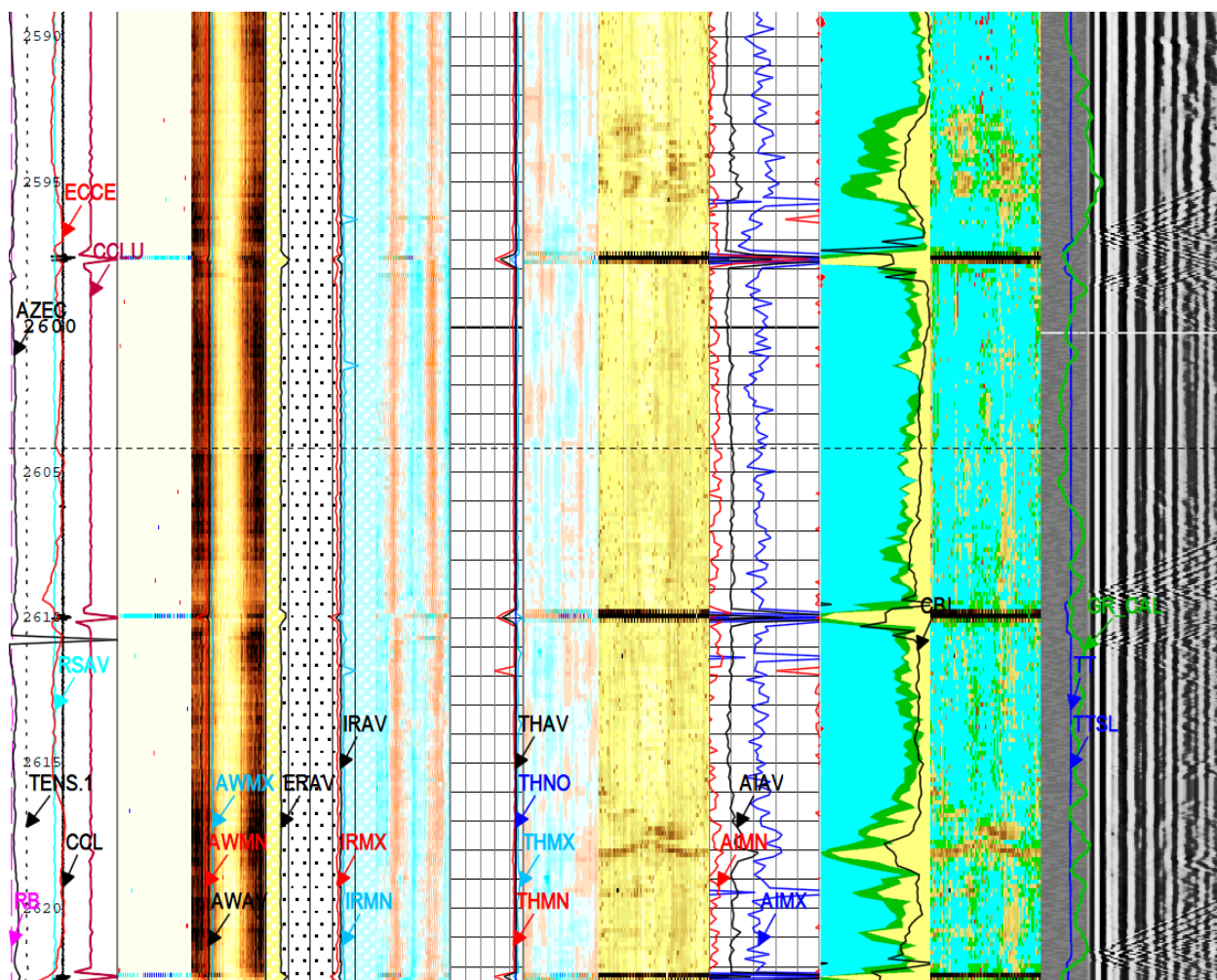


Figure 6: Example Log Section from 2590-2622m showing predominantly free pipe with very early signs of some weakly bonded formation material.